



Abstract Book

7TH DSEV ANNUAL SYMPOSIUM 2026
AT FACULTY OF HEALTH AND MEDICAL SCIENCES
COPENHAGEN UNIVERSITY

7th ANNUAL SYMPOSIUM 2026



Program

Thursday 1st of October 2026

Margrethe Bohr Salen, Niels Bohr Building,
Copenhagen University, Jagtvej 128, 2200 København N

09.00 – 10.00	Registration and breakfast
10.00 – 10.15	Welcome
10.15 – 11.15	Session 1: (Chair Per Svenningsen) Keynote Speaker: Edit Buzas, Dep. of Genetics, Cell- and Immunobiology, Semmelweis University, Hungary “Inducible changes in EV biogenesis”
11.15 – 11.35	Coffee break
11.35 – 12.30	Commercial sponsor session — Flash talk competition (Chair Aida Solhøj Hansen)
12.30 – 13.15	Joint photo and lunch
13.15 – 14.00	Poster session
14.00 – 15.00	Session 2: “EV characterization and therapeutics” (Chair Johann Mar Gudbergsson) Josefine B. Jensen, University of Southern Denmark: “Cancer-derived EVs in estrogen receptor positive breast cancer mouse models” Sofie Traynor, University of Southern Denmark: “The role of cancer-derived extracellular vesicles in therapy resistance” Camille M. Vandenbroucke, Aarhus University: “Tracing the Protein Corona Footprint of Extracellular Vesicles Mapped In Situ by EV-SPEC in Cells and Zebrafish” Stefan Vogel, University of Southern Denmark: “Cargo loading of extracellular vesicles by programmable fusion with synthetic liposomes”
15.00 – 15.30	Coffee break
15.30 – 16.30	Session 3: “EV Biomarkers” (Chair Javier Donoso-Quezada) Katja L. Cliff, Aalborg University Hospital: “Development of tissue-specific EV-based diagnostic tools for Metabolic Dysfunction -Associated Steatosis Liver Disease.” Mohamed Helal, Aarhus University: “MicroRNA biomarker discovery in depressed patients using the NanoString nCounter technology” Camilla M. T. Andersen, Roskilde University: “EV Characterization in Myeloproliferative Neoplasms Reveal Distinct Thrombo-inflammatory Profiles Among Patients vs Healthy Controls” Jonas K. Hemmingsen, Aarhus University: “Extracellular Vesicle-Associated Syndecan-1 Differentiates Pediatric Brain Tumor Patients with High-Grade from Low-Grade Pilocytic Astrocytoma”
16.30 – 16.40	Experience from last year’s MOVE Fellowship
16.40 – 17.30	General assembly, oral and poster presentation awards, MOVE award
18.15 –	Social Dinner in Madklubben Østerbro, Østerbrogade 79, 1 sal, (~25min walk from the venue)



Cancer-derived EVs in estrogen receptor positive breast cancer mouse models

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Estrogen receptor positive (ER+) breast cancer accounts for most breast cancer cases. Cancer progression is strongly influenced by interactions between tumor cells and the tumor microenvironment (TME), in which extracellular vesicles (EVs) play a key role. EVs are heterogeneous, nanosized membrane-enclosed particles carrying cargo such as proteins. This study aims to unravel the heterogeneity of EVs to understand how cancer-derived EVs communicate in two ER+ breast cancer models.

Murine ER+ breast cancer models B6BC and E0771 were established in female C57BL/6 mice. ER expression was evaluated in vivo by immunohistochemistry, and immune cell landscapes were characterized using spectral flow cytometry. EVs were isolated from conditioned cell media from cell lines by size exclusion chromatography and characterized by western blotting, NTA, nano flow cytometry and quantitative mass spectrometry.

ER expression was detected in E0771 and B6BC tumors and baseline immune profiling revealed distinct TME compositions. E0771 and B6BC-derived EVs expressed canonical EV markers. Proteomics profiling revealed distinct protein signatures, indicating heterogeneity in EV cargo.

Unraveling EV heterogeneity may provide deeper insights into mechanisms of immune communication in ER+ breast cancer.

The role of cancer-derived extracellular vesicles in therapy resistance

Sofie Traynor¹, Pernille G. Pedersen¹, Carla Alves¹, Henrik J. Ditzel¹, Aida S. Hansen¹

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Breast cancer is the leading cause of cancer-related death among women worldwide. Although current treatments with combined endocrine therapy and CDK4/6 inhibition have improved survival for patients with estrogen receptor-positive (ER+) breast cancer, most patients inevitably develop resistance to this treatment and experience disease progression. Cancer progression is strongly influenced by interactions between cancer cells and the tumor microenvironment (TME), with extracellular vesicles (EVs) emerging as important mediators of intercellular communication. Increasing evidence suggests that EVs contribute to therapeutic resistance across cancer types, in part by promoting immunosuppressive mechanisms. However, knowledge of EV subtype diversity and functional specialization of distinct EV populations is poorly understood. To address this knowledge gap, we applied single-EV surface proteomic profiling to resolve the heterogeneity of EVs secreted from a human breast cancer cell line with and without acquired resistance to endocrine therapy and CDK4/6 inhibition. Our findings show increased EV secretion by therapy resistant breast cancer cells and highlight a striking degree of heterogeneity of secreted EVs from both cell lines, with indications of specific markers and EV subpopulations being enriched in EVs from resistant cells. Future studies will address how specific EV-subsets contribute to immune evasion and treatment response in ER+ breast cancer.

Tracing the Protein Corona Footprint of Extracellular Vesicles Mapped In Situ by EV-SPEC in Cells and Zebrafish

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The biomolecular corona of extracellular vesicles (EVs) is commonly studied by physical isolation, adapted from the synthetic nanoparticle field where the hard corona is characterized as a surface-bound complex. Whether it sufficiently captures the corona formed on EVs, however, remains unclear. We developed EV-SPEC, a proximity labelling approach enabling in situ mapping of EV surface-interacting proteins from biogenesis to post-secretion. EV-SPEC uses TurboID, which biotinylates proteins within a few nanometer distance from the fused bait protein. CD63 was selected as the bait due to its ease of genetic EV engineering. Multiple CD63-TurboID variants enabled spatially resolved labelling and discrimination between proteins associated with the EV surface and luminal cargo. In vitro EV-SPEC identified typical bovine proteins such as “sticky” serum proteins previously reported in nanoparticle and EV hard coronas. Notably, human intracellular proteins involved in vesicle biogenesis and cell architecture were also identified, potentially reflecting EV surface interaction footprints in multivesicular bodies and amphisomes. We have further developed in vivo EV-SPEC in zebrafish to map EV corona proteins in situ which we confirmed association of biotinylated corona proteins with EVs in the blood. EV-SPEC therefore provides a complementary approach to physical isolation, revealing the dynamic and composite nature of the EV corona.

Cargo loading of extracellular vesicles by programmable fusion with synthetic liposomes

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Extracellular vesicles (EVs) hold great potential as natural nanocarriers for drug delivery applications, yet effective, gentle cargo-loading strategies for EVs remain a major challenge. We present a fully programmable liposome-EV fusion platform that is controlled by hybridization of lipidated nucleic acids (LiNAs). LiNA technology enables controlled, programmable, multi-step cargo loading from liposomes into the EV lumen under mild conditions. LiNAs enable highly efficient fusion for different cargos, such as oligonucleotides, proteins, and small molecules, with low leakage. We demonstrate both single-step and multi-step cargo transfer by LiNA-mediated fusion between liposomes and EVs, creating EV hybrids (EVHs). EV hybrid (EVH) combinations containing different EV types and liposome populations with different cargos were prepared, and transfer of protein cargo was demonstrated by the separate delivery of β -Gal (>450 kDa) and its substrate, quantified by release of fluorescein into the EV hybrid (EVH) lumen. Cryo-TEM captured early- and late-stage fusion states between EVs and liposomes, showing distinct membrane interactions between complementary LiNA-encoded populations. In summary, the reported results position LiNA-mediated fusion as a controlled, programmable strategy for sequential engineering EVs, opening opportunities for advanced nanocarrier designs based on EV hybrids (EVHs).

Development of tissue-specific EV-based diagnostic tools for Metabolic Dysfunction-Associated Steatosis Liver Disease

Katja Lund Cliff^{1,2}, Javier Donoso Quezada¹, Rikke Bæk³, Rikke Wehner Rasmussen¹, Rikke Bülow Eschen¹, Morten Hjuler Nielsen¹, Malene Møller Jørgensen^{2,3}, Aase Handberg^{1,2}, Maiken Møllergaard^{1,2}

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Metabolic dysfunction-associated steatotic liver disease (MASLD) is the leading cause of chronic liver disease, lacking applicable screening tools. Extracellular vesicles (EVs) have shown great biomarker potential with reported involvement in MASLD progression. We recently showed increased levels of circulating liver-derived (ASGR1+) EVs in individuals with MASLD compared with an obesity control group. Moreover, screening for EV-expressed biomarkers via our Next Generation Diagnostics (NGDx) platform, we found altered intensities of several promising markers, including CD36, FATP1, and TREM2 in MASLD using EV Array.

Using high-resolution flow cytometry (hFCM), the aim of this study is to determine co-localization and plasma concentrations of these EV phenotypes for comparison between three study groups: Individuals with obesity and MASLD (n=36) before and during weight loss intervention, obesity control group (n=25), and lean control group (n=27). Results will further be related with clinically relevant parameters, including liver fat content. At present, two different four-marker panels for hFCM have been validated using obesity plasma pools. Early preliminary findings indicate decreased levels of TREM2+-EVs and increased levels of FATP1+CD36+-EVs in obesity.

With quantification of these two panels, we take the next step towards advancing the development of circulating EV profiles as biomarkers for MASLD.

MicroRNA biomarker discovery in depressed patients using the NanoString nCounter technology

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Major depressive disorder (MDD) is a leading cause of disability worldwide and remains a major clinical challenge because diagnosis relies primarily on clinical symptoms, which may overlap with other psychiatric disorders, while antidepressant response is heterogeneous and often delayed. In Denmark, MDD affects approximately 3–7% of the population at any given time, with a lifetime prevalence of 17–18%. Circulating microRNAs (miRNAs), key post-transcriptional regulators of gene expression, have emerged as promising minimally invasive biomarkers for MDD and treatment response, although previous findings remain inconsistent. In this Danish cohort, age- and sex-matched plasma samples from patients with MDD collected before and after 8 weeks of escitalopram treatment and from healthy controls were analyzed using the human NanoString nCounter panel covering 827 miRNAs. Following optimization of plasma RNA extraction and purification, a high standard extraction protocol was developed with consistent computational downstream analysis. Preliminary analyses identified miRNA signatures associated with MDD disease specific and antidepressant response markers, including sex-specific expression patterns. These findings support circulating miRNAs as candidate biomarkers for diagnosis, treatment stratification, and earlier assessment of therapeutic response, with potential to advance precision psychiatry.

EV Characterization in Myeloproliferative Neoplasms Reveal Distinct Thrombo-inflammatory Profiles Among Patients vs Healthy Controls

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The chronic leukemias, Myeloproliferative neoplasms (MPN) are a result of somatic driver mutations (JAK2V617F/CALR/MPL) leading to constitutive activation of the JAK-STAT pathway. The consequences are elevated blood cell and platelet counts, chronic inflammation and death from cardiovascular events (CVE). Further investigation into the mechanisms causing the persisting risk of CVE is needed, as even with intervention and normalized blood counts, the threat is not resolved within this patient group. We performed characterization of EV surface proteins using the EV Microarray platform. 40 antibodies against MPN associated and thrombo-inflammatory proteins were measured against 157 samples (MPN=120, control=37). The proteins were visualized by biotinylated anti-CD9, -63 and -81. The analysis showed no correlation between blood counts or EV protein expression, indicating separate mechanisms for EV protein distribution.

Results show that MPN EVs from patients harboring the JAK2V617F but not the CALR mutation, carry elevated levels of proteins associated with inflammation (IL1-b, IL6, IL8, IL10), immune signaling and vascular remodeling (CXCR4, TGFBR2) and platelet/endothelial activation (vWF, CD62P, CD36, GPIIb/CD61, TSP-1, PECAM-1). This compels further investigation of JAK2V617F cell derived EVs as potential amplifiers of inflammation and CVE in MPN.

Extracellular Vesicle-Associated Syndecan-1 Differentiates Pediatric Brain Tumor Patients with High-Grade from Low-Grade Pilocytic Astrocytoma

Jonas Klejs Hemmingsen^{1,2,3}, Josephine Elgaard Johansen⁴, Mads Zippor^{1,2}, Bradley Whitehead^{1,2}, Stine Mary Vissing⁵, Hannah Weissinger³, Mette Galsgaard Malle³, Samson Folami^{1,2,6}, Kenneth A. Howard³, Srinivas Gopala⁷, Lasse Sommer Kristensen⁵, Peter Nejsum^{1,2}, Torben Stamm Mikkelsen^{4*}, Vineesh Indira Chandran^{1,2,8*}

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Reliable non-invasive biomarkers for tumor grading and disease monitoring in pediatric brain tumors are an unmet clinical need. Circulating extracellular vesicles (EVs) carrying molecular cargo reflective of tumor biology, offer promise as liquid biopsy tools. In this exploratory study, plasma EVs were isolated from pediatric brain tumor patients (n=60) by size-exclusion chromatography and characterized using nanoparticle tracking analysis, cryo-electron microscopy, immunoblotting, and single-vesicle total internal reflection fluorescence imaging. EV-SDC1 was quantified and analyzed in relation to tumor grade, subtype, and tumor volume. EV-SDC1 levels were significantly elevated in high-grade compared with low-grade pilocytic astrocytoma tumors and robustly discriminated high- from low grade pediatric brain tumor patients with a diagnostic accuracy of 0.66. SDC1 mRNA expression in matching tumor tissues showed a similar discriminatory pattern (AUROC = 0.71). Importantly, EV-associated SDC1 showed substantially greater discriminatory potential than soluble plasma SDC1. Independent analysis of transcriptomic data from the Open Pediatric Brain Tumor Atlas further demonstrated differential SDC1 expression across pediatric brain tumor subtypes. Collectively, these findings support circulating EV-SDC1 as a potential biomarker of tumor severity and provide a rationale for its further evaluation in larger cohorts for non-invasive disease stratification and longitudinal monitoring.

POSTER PRESENTATIONS:

Poster no. 1:

Detecting tissue-specific microRNA secretion in hyperglycemia using bio-orthogonal RNA tracing and nano-sensors

Xiaohan Lin¹, Louise Torp Dalgaard¹, Pratik Shah¹, Anand Hardikar²

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² Medical School, Western Sydney University, Campbelltown, New South Wales, Australia

Extracellular RNAs (exRNAs) are promising biomarkers and mediators of intercellular communication, but their tissue origins and regulation during impaired glucose metabolism remain poorly understood. This project aims to identify tissue-specific exRNAs released under hyperglycemic conditions using a novel bio-orthogonal RNA labeling strategy, develop nano-sensors for their rapid detection, and validate candidate biomarkers in clinical samples from the Danish Childhood Obesity Biobank.

Using cell-specific BS2-mediated RNA labeling in ex vivo mouse and human tissues, newly synthesized and secreted RNAs will be tracked and characterized. Sensitive nano-sensors will be developed for quantitative exRNA detection and validated against RT-qPCR. Candidate biomarkers will be evaluated in children with obesity to determine their association with metabolic deterioration. Functional studies of selected exRNAs will be conducted in collaboration with Professor Anand Hardikar at Western Sydney University.

This research will advance understanding of exRNA secretion mechanisms and support the development of minimally invasive diagnostic tools for early detection and monitoring of metabolic disorders.

Poster no. 2:

Dietary patterns associate with cytokine cargo and inflammatory profiles of circulating extracellular vesicles in obesity-associated MASLD and during weight loss

Javier Donoso-Quezada¹, Randy Tobberup^{2,3}, Malwina Ulanowska^{1,4}, Rikke Wehner Rasmussen¹, Gunna Christiansen⁵, Maiken Møllergaard^{1,4}, and Aase Handberg^{1,4}

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Extracellular vesicles (EVs) transport bioactive molecules, including cytokines, and may contribute to inflammatory signaling in metabolic disease. Here, we investigated circulating EV-associated cytokines in obesity-related Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD) and during a weight-loss intervention, together with their relationships to serum cytokines, clinical parameters, and dietary patterns. EV-associated cytokine profiles differed from cytokines measured in serum, suggesting EV-specific inflammatory regulation. EV-associated IL-6, TNF- α , IFN- γ , and other cytokines showed strong associations with clinically relevant features of obesity and MASLD, including liver fat content, AST, and HbA1c. During weight loss, longitudinal trajectories in EV-associated IL-2, IL-4, TNF- α , and IL-1 β clustered with changes in liver fat and fibrosis-related indices, including MACK3 and FNI, indicating coordinated remodeling of inflammatory and metabolic features during the intervention. Moreover, dietary patterns were associated with the direction of change in IL-2, IL-1 β , IFN- γ , and TNF- α during weight loss. Together, these findings indicate that circulating EVs display a distinct inflammatory cytokine profile in obesity-associated MASLD and suggest that dietary patterns may be linked to changes in EV-associated cytokine cargo during weight loss. This highlights EVs as a potential interface between nutrition, metabolic dysfunction, and systemic inflammation.

Poster no. 3:

DRONE – Delivery of RNA Origami Nanoscaffolds via Exosome

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What if tiny homing nanomachines could patrol the body and converge at sites of injury? We present our progress towards DRONE, a strategy to harness the biology of extracellular vesicles(EVs) for packaged and targeted delivery of theranostic RNA origami. Using zebrafish as a model system, we show tissue-specific expression and loading of RNA into genetically engineered EVs which travel from donor to recipient cells in vivo. The journey of these EVs and their RNA cargo was simultaneously visualised by an EV-anchored pH-sensitive reporter protein and mRNA with or without the fluorogenic aptamer Pepper. We find efficient delivery and translation of the mRNA cargo in the recipient endothelial cells when coupled with the viral fusogen VSV-G, while the Pepper signal remains weak. Future studies will focus on improving RNA loading into EVs and enhancing Pepper stability through circularisation to advance the programable delivery of increasingly complex RNA structures, including RNA origami.

Poster no. 4:

Tracing the Fate of Glioblastoma-derived Extracellular Vesicles in a Xenograft Model

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Extracellular vesicles (EVs) mediate intercellular communication by transferring proteins, RNA, DNA, and lipids, thereby regulating differentiation, proliferation, and metastasis. Exosomes are central to cancer progression, yet tracking EV dynamics is limited, as conventional fluorescence methods cannot selectively label distinct EV populations over time. To overcome this, we designed a cd63-HaloTag construct containing the pH-sensitive marker pHluorin. HaloTag irreversibly binds a tetra-methylrhodamine-linker (TMR), allowing selective fluorescent labeling. Using Tol2 transgenesis in zebrafish, we applied a HaloTag pulse-labeling strategy with pHluorin, TMR, and the blocking ligand 7-bromoheptane (7BR), enabling time-resolved EV tracking. We observed accumulation patterns in vivo in the caudal vein.

Building on this system, we aim to trace tumor-derived EVs, particularly from glioblastoma, which drive tumor spread and pre-metastatic niche formation. Zebrafish embryos are an attractive host for xenografting human cancer cells due to their early lack of adaptive immunity. By transfecting glioblastoma cells with HaloTag constructs prior to xenografting into the zebrafish brain, we will monitor the spatiotemporal dissemination of glioblastoma-derived EVs (GEVs) via circulation. This model will provide insight into GEV-mediated tumor propagation and evaluate blood-borne GEVs as potential early biomarkers of glioblastoma.

Poster no. 5:

Longitudinal profiling of extracellular vesicle surface markers in blood and CSF after traumatic spinal cord injury

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Background: Traumatic spinal cord injury (TSCI) triggers complex secondary pathophysiological processes. Extracellular vesicles (EVs) in blood and cerebrospinal fluid (CSF) may serve as biomarkers for these neurocellular perturbations. This study investigates longitudinal changes in EV-surface markers in blood and CSF to elucidate how these responds to TSCI.

Objective: To characterize and compare the longitudinal expression profiles of EV-surface markers in blood and CSF following TSCI.

Methods: Fifteen TSCI patients and 15 non-TSCI patients were included. Blood and CSF were collected at median 0-, 9-, and 148- days post-injury. An advanced EV Array was used to target markers associated with nerves, inflammation, stress, mitochondria, and tight junctions.

Results: Multivariate expression patterns differed between blood and CSF. In CSF, significant differences were observed between TSCI and non-TSCI patients, but the EV-surface marker profile remained stable over time after injury in the TSCI group. Conversely, no significant differences were observed between TSCI and non-TSCI patients in peripheral blood but the EV-surface marker profile changed significantly over time after injury in TSCI patients.

Conclusion: TSCI elicits distinct EV-surface marker expression profiles in blood and CSF with different dynamics during follow-up. This suggest a compartmentalized EV-surface marker response to TSCI, implying that blood EV-surface marker profiles do not reflect that of CSF.

Poster no. 6:

Brain-Derived Extracellular Vesicles: A Promising Source of Blood Biomarkers for Traumatic Brain Injury

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Traumatic brain injury (TBI) is a major cause of mortality and long-term disability among the young, while current blood biomarkers remain limited by their temporal window and lack of brain specificity. Extracellular vesicles (EVs) may provide a more dynamic source of circulating molecular information, and brain-derived EVs could offer increased tissue specificity.

This project aims to establish a robust strategy for the enrichment of brain-derived EVs from plasma and to investigate their potential as blood-based biomarkers of TBI.

Plasma samples from patients with severe TBI and trauma controls without brain injury were pooled within groups. EVs were isolated by size-exclusion chromatography (SEC) and characterized by nanoparticle tracking analysis (NTA) and high-resolution flow cytometry (hFCM). EV-associated events were assessed using classical EV markers, including CD9, CD81 and lactadherin. Candidate brain-associated surface markers are being evaluated using hFCM and immunocapture to identify targets suitable for specific and reproducible enrichment of brain-derived EVs.

Preliminary analysis of pooled plasma showed higher particle concentrations by NTA and increased EV-associated events by hFCM in the TBI pool compared with trauma controls. Marker-positive events were approximately 2.4-fold higher in the TBI pool.

Establishing a reliable brain-EV enrichment strategy will enable subsequent characterization of EV cargo to identify TBI-associated molecular signatures.

Poster no. 7:

Quantifying longitudinal variation in circulating extracellular vesicles

Kirstine Kløve-Mogensen¹, Rikke Bæk¹, Rikke W. Rasmussen², Evo K. L. Søndergaard¹, Aase Handberg^{2,3}, Maiken Møllergaard^{2,3}, Malene M. Jørgensen^{1,3}

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Background: The clinical potential of extracellular vesicles (EVs) as biomarkers is limited by a lack of characterized baseline variation. Natural fluctuations in EV concentration and phenotype among healthy individuals pose a significant challenge to the reproducibility of biomarker discovery.

Objective: To characterize the intra-individual and inter-individual longitudinal dynamics (day-to-day and week-to-week) of circulating EVs in healthy donors using a multi-platform approach.

Methods: Blood samples were collected from four healthy female donors over five consecutive days and six weeks. To ensure comprehensive characterization, EVs were analyzed using Nanoparticle Tracking Analysis (NTA), high-resolution flow cytometry (hFCM) for three surface markers, and an EV Array covering 23 markers. Correlative analysis with blood cell counts was also conducted.

Results: Data reveals significant fluctuations in both EV abundance and phenotypic marker expression. While CD9+ EVs were the predominant population across all platforms, CD63+ and CD81+ populations exhibited markedly higher variability. Notably, long-term variation was more pronounced than short-term (daily) changes.

Conclusion: These findings demonstrate that EV profiles in healthy populations are highly dynamic. To mitigate the risk of misinterpretation in clinical studies, it is essential to establish standardized reference ranges and harmonize methodologies to acknowledge for this inherent biological variability.

Poster no. 8:

Metabolic and immune system markers are increased on circulating extracellular vesicles in carriers of m.3243A>G mitochondrial disease

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Background/objectives: m.3243A>G carriers frequently exhibit a low body mass index, suggesting altered systemic metabolic activity. Metabolic activity is closely linked to immune activation, yet the mechanisms coordinating metabolic and immune processes in these individuals are incompletely understood. Extracellular vesicles (EVs) may serve as informative biomarkers for systemic processes. We therefore aimed to characterize metabolic substrate transporters and immune response markers on circulating EVs.

Methods: EVs from plasma were phenotyped in 24 m.3243A>G carriers and 24 age- and sex-matched healthy controls using EV Array. A panel of antibodies targeting 13 glucose and fatty-acid transporters, 15 immune response markers, and four vesicle-related markers were applied.

Results: EVs from m.3243A>G carriers showed increased expression of glucose transporter type 4 (GLUT4) (p=0.03), long-chain fatty acid transporter proteins 1 (FATP1) (p=0.01), and 4 (FATP4) (p=0.04), as well as immune markers MHC class I chain-related protein (MIC)A/B (p=0.04), cluster of differentiation (CD)45 (p=0.03), and the vesicle-related CD81 (p=0.01).

Conclusions: Increased GLUT4, FATP1, and FATP4 on EVs from m.3243A>G carriers may indicate altered systemic metabolic. In parallel, elevated EV expression of MIC/A/B and CD45 could suggest changes in immune regulation, along with increased CD81, suggesting modified cell signaling processes.

Poster no. 9:

Cytoprotective Effect of Extracellular Vesicles from Conditioned Red Blood Cells During Stroke

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Acute ischemic stroke (AIS) is a leading cause of death worldwide. Remote ischemic conditioning (RIC) has emerged as a promising therapeutic strategy to reduce ischemic-reperfusion injury (IRI), potentially mediated by extracellular vesicles (Evs). This study aimed to investigate the cytoprotective effects of Evs derived from red blood cells (RBCs) subjected to in vitro RIC. Evs were isolated using size exclusion chromatography and characterized by nanoparticle tracking analysis and western blotting for EV (CD9, flotillin-1) and RBC (CD235a) markers. Fractions 3-5 were identified as EV-rich and used in functional assays. To assess the potential protective effects of RBC-derived Evs, immortalized human brain microvascular endothelial cells (IM-HBMECs) were subjected to oxygen-glucose deprivation (OGD), mimicking stroke-like conditions. ATP levels were measured post-OGD to evaluate metabolic activity. No statistically significant protective effect of RBC-derived Evs was observed. However, a trend suggesting differential effects of Evs under normoxic versus anoxic conditions was noted, with some indications of reduced ATP levels under normoxia.

These findings suggest that RBC-derived Evs may exert context-dependent effects on endothelial cells. Further optimization of experimental conditions and replication of results are necessary to clarify their therapeutic potential in stroke treatment.

Poster no. 10:

Through The Looking Glass: Mapping Endogenous Extracellular Vesicle Cargo Delivery in the Zebrafish Brain

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The concept of extracellular vesicles (EVs) mediating communication across cells and systems is often taken for granted. However, there are no *in vivo* studies demonstrating whether endogenous EV cargo escapes the endosome prior to degradation. It could be argued that only the EV's membrane-bound molecules (and biomolecular corona) need be functional: yet they contain many well-studied, important cargo, and the literature shows a plethora of significant differences in EV content caused by changes in their environment. To address this, we will package Cre recombinase inside the EVs of embryonic zebrafish brains. Successful endosomal escape by Cre will permanently switch fluorescent reporter expression in recipient cells from blue to red – evidence that EVs do deliver functional cargo *in vivo*. Specifically, we will overexpress a brain EV ‘hijack’: a transmembrane protein fused to an fluorescent marker at the extracellular face, and Cre recombinase tethered by a self-cleaving protein at the luminal face. This approach has several advantages: We will be easily able to image the path of these EVs, thanks to the transparency of embryonic zebrafish; and do so without disturbing any endogenous communication network. Combining our approach with transcriptomic analysis, we will then profile changes correlated with successful cargo delivery from our tagged EVs. This will create a single-cell-resolution atlas of intercellular communication, beyond synapses.

Poster no. 11:

Optimized extracellular vesicle enrichment for deep proteomic profiling of bone marrow plasma in acute lymphoblastic leukemia

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Extracellular vesicles (EVs) are implicated in bone marrow signaling and protection of leukemic blasts in acute lymphoblastic leukemia (ALL), but recovery of the EV proteome is method dependent. We evaluated three enrichment techniques: differential ultracentrifugation (UC), strong anion exchange (SAX) magnetic beads and size exclusion chromatography (SEC). Each was optimized independently, then compared at optimal conditions in triplicate using pooled bone marrow plasma from patients with ALL. Unprocessed plasma (Raw) was analyzed for comparison. Isolated EVs were digested and analyzed on an Evosep One (11-min gradient, 100 SPD) coupled to an Orbitrap Astral mass spectrometer operated in narrow-window DIA mode. UC yielded the deepest proteome, with a median of 6,715 identified proteins versus 4,641 (SAX), 3,649 (SEC) and 972 (Raw). Important EV markers (CD9, CD63 and CD81) were detected in all EV-enriched samples irrespective of method, but not in any Raw samples. Lipoprotein co-isolation persisted throughout enrichments. The leukemia- and niche-associated markers CD34, CD10, ITGA4 and PLXNB2 were identified by all three techniques, whereas CD19, CD22 and CXCR4 were recovered only by UC, indicating higher sensitivity for proteins of interest. Applied to nine diagnostic and nine remission ALL patients, the UC pipeline yielded 1,301 differentially abundant proteins and recovered a coherent B-ALL immunophenotype (CD19, CD10, CD34 and ITGA4) from the EV compartment alone.

Poster no. 12:

Effects of remote ischemic conditioning and blood flow-restricted resistance exercise on extracellular vesicle function in heart failure

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Introduction: Early reperfusion limits myocardial injury following infarction but contributes to ischemia-reperfusion injury (IRI), for which effective pharmacological prevention is lacking. Remote ischemic conditioning (RIC) and blood-flow-restricted resistance exercise (BFRRE) may induce systemic cardioprotection through circulating extracellular vesicles (EVs). However, their cardioprotective potential and molecular effects in heart failure (HF) remain insufficiently characterized.

Methods: Patients with HF underwent a single bout of RIC, BFRRE, or no intervention (NIC). Plasma was collected at baseline, 5 min, and 30 min post-intervention. EVs were isolated by size exclusion chromatography and tested in an in vitro IRI model using H9c2 cells exposed to oxygen-glucose deprivation. Cell viability was assessed by ATP-based bioluminescence, and EV-treated cells were analyzed by RNA sequencing.

Results: RIC-EVs at 30 min significantly increased viability compared to baseline RIC-EVs and NIC-EVs at 30 min, though the between-group difference in change over time was not significant. Transcriptomic analysis showed RIC-EV-associated responses under both normoxia and hypoxia, with shared and oxygen-modified responses varying by time point.

Conclusion: RIC, but not BFRRE, altered the viability-promoting effects of circulating EVs in vitro, supporting further investigation of RIC-EVs as mediators of systemic cardioprotection in HF, with time- and oxygen-dependent transcriptional effects.

Poster no. 13

Decoding the heterogeneity of T-cell derived extracellular vesicles

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Accumulating evidence suggests that extracellular vesicles (EVs) play important regulatory roles in a wide range of inflammatory diseases, including psoriasis. Psoriasis is a chronic inflammatory skin disease driven by a persistent inflammatory loop sustained in part by cytokines released from activated T cells. Recent studies have demonstrated that co-stimulatory proteins and cytokines are associated with EVs, raising the hypothesis that distinct EV subpopulations released from activated T helper cells may contribute to the transmission of inflammatory signals driving disease-associated inflammation.

To investigate activation-dependent heterogeneity of T-cell-derived vesicles, we used nano flow cytometry and Proximity Barcoding Assay (PBA) to profile the protein expression of EVs isolated from aCD3/aCD28 and IL-2 activated CD4⁺ T cells from healthy blood donors.

Activated T helper cells generated EV populations with altered scattering profiles, suggesting differences in EV size and composition compared with EVs from resting cells. PBA analysis identified 15 distinct EV subpopulations, several of which were enriched following T cell activation and exhibited increased expression of immune-related markers. Future studies will aim to characterize EV subpopulations derived from T cells in psoriatic and healthy skin, and define their roles in immune regulation, providing new insights into EV-mediated immune communication and inflammatory disease pathogenesis.

Poster no. 14:

Disrupting intercellular communication in Alzheimer's Disease: APOE-mediated alterations in glial extracellular vesicle signatures

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Alzheimer's disease (AD) is characterised by progressive neurodegeneration and chronic neuroinflammation. Although APOE4 is the strongest genetic risk factor for late-onset AD, the exact mechanisms linking APOE genotype to AD pathogenesis and progression remain incompletely understood. Extracellular vesicles (EVs) are nanosized, membrane-enclosed particles that mediate intercellular communication but may also propagate disease-associated signals. This project investigates how APOE genotype influences EV composition and signalling in microglia and astrocytes, two key regulators of brain homeostasis and neuroinflammation. Using isogenic human induced pluripotent stem cell-derived microglia and astrocytes expressing different APOE isoforms, EVs will be isolated under basal and AD-relevant conditions and characterised through proteomic and lipidomic profiling. Their functional effects will be assessed in a human iPSC-derived tri-culture comprising neurons, astrocytes and microglia, with single-cell RNA sequencing used to resolve cell type-specific responses. By integrating APOE-dependent EV signatures with recipient-cell responses, this work aims to establish a human-relevant framework for understanding how microglial and astrocytic EVs contribute to APOE4-associated neuroinflammation and may reveal new biomarkers and therapeutic targets in AD.

Poster no. 15:

Characterisation of circulating extracellular vesicles subpopulation following remote ischemic conditioning in acute ischemic stroke

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Remote ischemic conditioning (RIC) is being investigated as a therapeutic strategy for acute ischemic stroke, but the circulating mediators underlying its effects remain to be completely understood. Extracellular vesicles (EVs) are candidate mediators of systemic signalling and may reflect vascular, erythrocyte, and immune responses to ischemic injury and RIC. We aimed to characterise circulating EV populations before and after RIC in patients with acute ischemic stroke and to compare their profiles with a small healthy reference group. Paired plasma samples were collected from nine participants in the eNOS study before and 2 h after RIC. Plasma EVs were isolated and analysed by nanoscale flow cytometry using CD9/CD63/CD81 to identify tetraspanin-positive events and CD31, CD235a, and CD56 to define marker-positive subsets. Event concentrations were corrected for sample dilution, and changes were evaluated using paired analyses and log₂ fold changes. Preliminary analyses indicate substantial interindividual variation, without a uniform short-term change across the investigated EV populations. Comparison with healthy reference samples will provide additional context for population-specific differences. These findings support further investigation of circulating EV heterogeneity and temporal responses to RIC in acute ischemic stroke.

Poster no. 16:

Phenotype of Circulating Pathogenic Extracellular Vesicles in Minimal Hepatic Encephalopathy

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Minimal hepatic encephalopathy (MHE) is present in 40% of patients with cirrhosis, who exhibit cognitive and motor impairment. MHE associates with a shift in peripheral inflammation, and reversal of this shift with rifaximin improves MHE. However, how peripheral alterations are transmitted to the brain to induce cognitive impairment remains unclear. Plasma extracellular vesicles (EVs) from patients with MHE are pathological and play a role in the induction of cognitive impairment. Given that EVs are released by most cells, characterizing how their profiles change under MHE might reveal relevant biomarkers linked to disease progression. The aims are to 1) identify and quantify liver (ASGR1+) and monocyte-derived (CD14+) plasma EVs in cirrhotic patients; 2) assess if CD36, up-regulated in MHE, varies in specific cell type EVs; and 3) determine if the observed changes can be related to the onset or reversal of MHE.

We obtained plasma samples from cirrhotic patients without MHE, patients with MHE, and patients with MHE treated with rifaximin. Surface-marker profiles of circulating liver- and monocyte-derived EVs will be characterized by high-resolution flow cytometry. Findings will be correlated with relevant clinical parameters to explore potential associations between EV-profiles and cognitive dysfunction, or MHE reversal. If validated, these changes could provide accessible biomarkers to anticipate MHE onset and assess treatment response to rifaximin in the future.

Poster no. 17:

Chronic high glucose alters tRNA-derived fragments in extracellular vesicles released by pancreatic β -cells

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tRNA-derived fragments (tRFs) are emerging regulators of pancreatic β -cell function and are altered during diabetes and metabolic stress. tRFs also participate in extracellular vesicles (EV) mediated intercellular communication involving β -cells. We investigated whether chronic high-glucose exposure which models glucotoxic stress associated with hyperglycemia alters the tRF cargo of β -cell-derived EVs. EVs were isolated by ultracentrifugation from conditioned media of rat insulinoma cells (INS-1E) cultured under low glucose (LG) and high glucose (HG). A low glucose plus mannitol osmotic control (LGM) was also used to distinguish glucose-associated effects from those caused by increased osmolarity. After characterization, EV RNA was analyzed by small RNA sequencing. Reads were hierarchically aligned to small RNA reference datasets including mature tRNA coordinates. tRFs represented the largest mapped RNA class in β -cell EVs. We identified several tRFs differentially represented between glucose conditions. Notably, a Glu-CTC 5' half (nt 1–35) was enriched in HG compared with both LG ($\log_2FC=3.26$, $FDR=0.0019$) and LGM ($\log_2FC=3.68$, $FDR=0.00091$). Additional glucose-associated fragments originated from Gly, Ala, Cys, Asp, Lys and Val tRNAs. Together with emerging evidence that tRFs regulate β -cell responses to metabolic stress and can participate in EV-mediated intercellular communication, these findings identify EV-associated tRFs as potential molecular signals of β -cell stress.

Poster no. 18:

Circulating microRNAs modulate tumor immunity and correlate with CNS tumor status

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Neutrophils are increasingly recognized as important contributors to tumor progression and therapy resistance, while microRNAs (miRs) mediate their interaction with the tumor microenvironment. Among these, miR-25 and miR-10b have emerged as oncogenic miRs involved in tumor progression and immune modulation. We profiled circulating miRs in plasma-derived extracellular vesicles (EVs) from patients with primary and secondary central nervous system tumors (n = 60; low- and high-grade glioma, cerebral metastasis) and healthy donors (HD, n = 10). In peripheral neutrophils from glioblastoma patients (n = 10), we analyzed NET formation, PD-L1 expression and miR profiles. Furthermore, HD-derived neutrophils were co-cultured with glioblastoma EVs with or without miR-25 inhibition to assess effects on neutrophil function. EV-associated miR-25 and miR-10b were significantly elevated in glioblastoma and cerebral metastases compared with low-grade glioma and HD (p < 0.0001). Their entity-specific signatures showed stronger discriminatory associations with tumor stage. Neutrophils from glioblastoma patients exhibited increased NET formation and PD-L1 expression compared with HD (p < 0.0001). Glioblastoma EVs enhanced NET formation in HD neutrophils, an effect markedly reduced by miR-25 inhibition.

These findings identify both miRs as promising diagnostic and therapeutic targets and implicate miR-25 in neutrophil-mediated immune modulation, providing a strong basis for further investigation.

Poster no. 19:

Engineering Antigen Presenting Extracellular Vesicles from THP-1 Cells for Cell Free Nano-Immunotherapy

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Dendritic cell (DC) based immunotherapies elicit strong antigen-specific responses, but clinical use is limited by complex production and instability. Extracellular vesicles (EVs) offer a stable, cell-free alternative retaining key immunological features of parent cells. Using lentiviral transduction, we engineered a THP-1 monocytic line stably expressing a single-chain construct linking the immunodominant CMV pp65 peptide to HLA class I/β2-microglobulin, plus CD80/CD86, generating antigen-presenting EVs for antigen specific T-cell activation.

Flow cytometry confirmed engineering success: HLA-ABC was consistently high, and CD80/CD86 were expressed in >99.9% of transduced cells (MFI ~31,000 and ~60,000) versus 3.3% and 1.7% in wild-type (WT) cells ($p < 0.0001$). Transduced and WT THP-1 cells were differentiated into immature/mature DCs (GM-CSF, IL-4, TNF-α, ionomycin) and M0 macrophages (PMA). Maturation markers CD40, CD83, CCR7, and HLA-DR increased progressively, confirming a mature antigen presenting phenotype.

EVs isolated by differential ultracentrifugation from WT and transduced cells across all differentiation states showed mean diameters of 110-195 nm and concentrations of 1.14×10^8 - 1.24×10^{11} particles/mL, consistent with small EVs.

This establishes a standardized engineered THP-1 platform for antigen-presenting EV production. Ongoing work will characterize the EV protein corona and evaluate CD8⁺ T-cell activation to assess therapeutic potential.

Poster no. 20:

Lipidomic characterization of extracellular vesicles in plasma from individuals with obesity and metabolic dysfunction-associated steatotic liver disease

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Introduction: Metabolic dysfunction-associated steatotic liver disease (MASLD) has rapidly become the most common chronic liver disease, affecting approximately 35-40% of the global adult population. The potential of extracellular vesicles (EVs) as biomarkers has already been linked with MASLD. Investigating EV content, especially lipid composition, can be a useful approach for identifying potential non-invasive biomarkers of MASLD.

Methods: The study included three groups: individuals with obesity and MASLD (n=36), obesity control group (n=25), and lean control group (n=27). Plasma EVs were enriched using the Mag-Net method and composition of lipids extracted from the EV-preparations was analyzed using LC-MS/MS.

Results: Distinguishing individuals with obesity and high liver fat from low liver fat, using a multivariate biomarker analysis model, revealed high discriminative power with AUROC of 0.96 with 100% sensitivity (CI: 74 - 100%) and 79% specificity (CI: 58 - 100%). The top five contributing lipids for the distinction included CL 64:5, SM 36:2, OAHFA 18:0/16:2, CL 64:10, HexCer 26:1. Interestingly, CL 64:5, SM 36:2, and CL 64:10 also significantly correlated with liver fat content.

Summary/Conclusion: This exploratory biomarker analysis identified individual lipid species with biomarker and correlative potential, suggesting that EV-lipid composition may be used as a diagnostic tool in MASLD.

Poster no. 21:

Proteomic profiling of extracellular vesicles from CHMP2B-mediated frontotemporal dementia patient-derived cells

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Frontotemporal Dementia (FTD) is a progressive neurodegenerative disorder caused by neuropathological changes in the frontal and temporal lobes. Familial forms of FTD cover 30-40% of all cases, where a pathogenic variant in the charged multivesicular body protein 2B-gene (CHMP2B) causes a rare form of familial FTD, FTD-3, affecting behavior and executive functions. CHMP2B is part of the endosomal sorting complex required for transport (ESCRT-III), which is important for the endo-lysosomal system and in formation of multivesicular bodies (MVBs) in cells. Knowing that exosomes originate from MVBs, we hypothesized that the CHMP2B mutation may influence the proteome of extracellular vesicles (EVs), and that the EVs could be involved in propagating the disease between brain cells. We made a proteomic profiling of EVs isolated from induced pluripotent stem cell (iPSC)-derived microglia and neurons obtained from FTD-3 patients. The EV proteomes will be compared to gene corrected controls and healthy donors. Additionally, the secreted non-EV fraction and on the whole cell lysates will also be analyzed to obtain a comprehensive picture of the disease-causing CHMP2B mutation. Focus is on differentially regulated proteins, which could drive the pathology, and comprise potential therapeutic targets. The study will provide new insight into the role of CHMP2B by addressing its impact on the proteome of disease-relevant cells and EVs in FTD-3.

Poster no. 22:

Effect of storage temperature and duration on extracellular vesicle's stability

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Extracellular vesicles (EVs) have been shown to contribute to physiological and pathophysiological processes and are therefore of high interest for potential therapeutic and diagnostic uses. Blood EVs are of particular interest as they can be obtained through a minimally invasive procedure and originate from blood cells, endothelial cells, and other tissues throughout the body. Consequently, circulating EVs can reflect changes in multiple organs and tissues, making them attractive as biomarkers for the diagnosis, monitoring, and treatment of a diverse group of diseases. However, EVs are sensitive to environmental conditions, and the optimal storage conditions to maintain their integrity have not yet been agreed upon. Here, we aim to explore how different storage temperatures and durations affect human plasma EV size and concentration. EVs were purified from plasma from two male donors using size exclusion chromatography (qEV10, 70 nm, Izon). The samples were analysed on the day of purification and again after 1, 7, and 14 days of storage at either 4°C, -20°C, or -80°C. EV concentration and size (nanoparticle tracking analysis (NTA)) and protein concentration (OD280) were measured. Lastly, we performed Western blotting using antibodies against standard EV proteins.

Long-term storage of EVs is essential for exploring the diagnostic potential in plasma samples, as analysis of clinical blood samples is often not possible immediately after isolation.

Poster no. 23:

Recipient cell identity governs intracellular transport and membrane interactions of extracellular vesicles across species

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Synthetic nanoparticles such as liposomes and lipid nanoparticles (LNPs) are widely used for drug delivery but are often associated with toxicity and immune responses. Natural extracellular vesicles (EVs) offer a promising, biocompatible alternative. However, almost all cells produce EVs, and there is currently limited knowledge of how different EV origins influence their uptake and intracellular behavior. Quantitative microscopy is emerging as a powerful tool to study these processes, using single-vesicle tracking in live cells to bypass ensemble averaging.

EVs are isolated from C2C12 cell culture, human serum, mouse plasma, and bovine milk, and are characterized following the MISEV 2018 guidelines. To quantify delivery, we electroporate Cy5-labeled siRNA into EVs and image live cells using a confocal microscope. Single-vesicle tracking is performed on EVs with a fluorescently labeled membrane using HILO microscopy and analyzed with an in-house-developed Python script.

Quantification of siRNA uptake using single snapshots acquired by confocal microscopy shows similar delivery across EV origins. However, tracking EVs in live cells allows direct observation of their dynamic behavior, revealing distinct spatiotemporal patterns that are primarily driven by the recipient cell type rather than EV origin. We observe that EVs in HEK293-H cells are significantly more dynamic, and their behavior is more homogeneous compared to EVs in bEnd.3 cells.

Poster no. 24:

Milk extracellular vesicles modulate molecular pathways in gut cells during early life

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Milk contains miRNAs enclosed in extracellular vesicles (EVs), which may contribute to gut and immune maturation in newborns. Compared with human milk, infant formula is associated with increased risk of intestinal pathologies, particularly in preterm infants. Given the similarity between human and bovine milk miRNA profiles, we hypothesize that bovine milk EVs exert gut-protective effects across species. However, formula production substantially reduces EVs and associated miRNAs. The GutBioMod project investigates whether supplementation with bovine milk-derived EVs and miRNAs can promote gut development. EV uptake and biodistribution were studied using fluorescently labelled bovine EVs in an intestinal loop model in term-born piglets. EV-associated fluorescence was clearly detected in intestinal cells, supporting uptake by the intestinal epithelium.

In a subsequent intervention study, piglets were fed EV-containing fractions, with or without a milk matrix, for 24 hours. Intestinal tissues were analysed by bulk and single-nucleus RNA sequencing. EV supplementation induced significant changes in intestinal cell-type composition and transcriptional pathways, demonstrating a substantial biological response.

Together, these findings support milk EVs as bioactive components capable of modulating intestinal cellular programs and potentially contributing to the beneficial effects of maternal milk during early life.